

Dhruvil Thakor

Software Engineer | Distributed Systems & Backend Infrastructure

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EDUCATION

DA-IICT (Dhirubhai Ambani Institute of ICT)

B.Tech in Information and Communication Technology (ICT) | **CGPA: 7.7**

Gandhinagar, India

Oct 2021 – May 2025

TECHNICAL SKILLS

- **Languages & Databases:** Java, C++, Go, PL/SQL, PostgreSQL, DynamoDB, Cassandra, Redis
- **Core Competencies:** Distributed and Scalable Systems, Artificial Intelligence (AI), OOP, RAG, Microservices, LLM
- **Frameworks & Libraries:** Spring Boot, React, JPA/Hibernate, Mockito
- **Tools & Infrastructure:** Git, Docker, Kubernetes, Terraform, CI/CD, Kafka, Spring AI, LangChain4j, Pinecone, LLM APIs (OpenAI, Anthropic Claude, Groq), AWS

PROFESSIONAL EXPERIENCE

Barclays

Software Engineer

Pune, India

Aug 2025 – Present

- **Spearheaded enhancements** across 5 backend **Microservices (Java/Spring Boot)**, utilizing robust **System Design** to build scalable components supporting **100,000+** daily transactions.
- **Streamlined deployments** by integrating continuous **CI/CD** pipelines, **Kubernetes** orchestration, and AWS provisioning via **Terraform**, reducing manual configuration time by **40%**.
- **Optimized data processing** utilizing **SQL** within PostgreSQL, streamlining payload transformations and reducing average request latency by **15%**.
- **Led Root Cause Analysis (RCA)** and rigorous debugging for production incidents, maintaining **99.9%** uptime and reducing MTTR by **30%** through advanced APM log analysis.
- **Collaborated** within a **6-member** Agile/Scrum team to gather scope constraints and estimate work, successfully delivering **3** major data representation reports ahead of schedule.

Barclays

Software Engineering Intern

Pune, India

May 2024 – Jul 2024

- **Developed** scalable data ingestion pipelines to parse, process, and publish over **100,000** records utilizing **SQL** and message queues, eliminating manual processing bottlenecks.
- **Executed Unit Testing** via **Mockito** and performed component debugging during legacy system decoupling, accelerating API response times by **20%** while ensuring data integrity.
- **Participated** in **4+** weekly technical design discussions to plan and execute tasks aimed at improving the team's operational development ecosystem efficiency by **25%**.

PROJECTS

DocQA: Retrieval-Augmented Generation Engine | Java 21, Spring Boot, Spring AI, Ollama

- **Constructed** a local **RAG** ingestion pipeline using **Spring AI** and **Ollama**, leveraging Apache Tika and TokenTextSplitter to chunk **50+ page** documents into a **Vector Store** for context-aware LLM querying.
- **Implemented** a real-time streaming architecture via **Server-Sent Events (SSE)**, reducing Time-to-First-Token (TTFT) by **45%** for users interacting through the Thymeleaf UI.

hraftd: Distributed Key-Value Store | Go, HashiCorp Raft, BoltDB, HTTP API

- **Architected** a fault-tolerant, distributed key-value store in **Go** utilizing the **HashiCorp Raft** consensus algorithm, achieving strict leader election across **5-node** clusters with sub-10ms latency.
- **Integrated** persistent log storage using **BoltDB** with scheduled snapshotting, and exposed a RESTful HTTP API for seamless node clustering, handling up to **5,000** reads/sec.

High-Performance HTTP Proxy & Load Balancer | C++17, POSIX, Multithreading

- **Programmed** a multithreaded Layer-7 HTTP reverse proxy from scratch in **C++17**, utilizing non-blocking I/O polling and a custom thread pool capable of processing **10,000+** concurrent connections.
- **Designed** real-time routing algorithms (Round-Robin, Least Connections) and continuous TCP health checks, optimizing socket lifecycles via RAII to improve load distribution efficiency by **25%**.